# From the Fano plane to $E_8$: $F_2$ geometry and exceptional symmetry

**The projective spaces over $F_2$ encode the combinatorics of exceptional Lie groups with striking precision.** The 27 lines on a cubic surface (arxiv) are literally the 27 points of an elliptic quadric $Q^-(5,2)$ in PG(5,2), the Fano plane's 7 points dictate octonion multiplication (arxiv) and thereby generate $E_8$, and binary error-correcting codes over $F_2$ build the $E_8$ and Leech lattices. These are not analogies — they are exact mathematical identifications, where the Weyl groups of $E_6$, $E_7$, $E_8$ turn out to be orthogonal and symplectic groups over $F_2$, (arxiv) and the Freudenthal magic square construction works uniformly over any field (arxiv) including $F_2$ to produce split Chevalley groups of exceptional type.

## Two quadrics in PG(5,2) govern $E_6$ and $E_7$

The projective space PG(5,2) has **63 points** (the nonzero vectors of $F_2^6$ modulo scalars). Inside it sit two fundamentally different non-degenerate quadrics, and each connects to a different exceptional group.

The **hyperbolic (Klein) quadric $Q^+(5,2)$** has **35 points** on it and **28 points** off it. (MDPI) The 35 on-quadric points parametrize the lines of PG(3,2) via the classical Klein correspondence. (Wikipedia) The 28 off-quadric points (Frans Marcelis) correspond naturally to 2-element subsets of an 8-element set (since C(8,2) = 28), (MDPI) forming the **triangular graph T(8)** with strongly regular parameters srg(28,12,6,4). The complement graph has parameters srg(28,15,6,10). These 28 points correspond to the **28 bitangents of a smooth quartic plane curve**, (arxiv) and their symmetry group is the Weyl group $\mathbf{W(E_7) \cong Sp(6,F_2) \times \mathbb{Z}/2\mathbb{Z}}$, of order **2,903,040**.

The **elliptic quadric $Q^-(5,2)$** has **27 points** on it and **36 points** off it. This is where the 27 lines live. The identification is exact: the **27 isotropic points of $Q^-(5,2)$ are the 27 lines on a smooth cubic surface**. (Tue) Two isotropic points are collinear in the polar space (i.e., joined by a totally isotropic line) precisely when the corresponding lines on the cubic surface meet. The full automorphism group of this configuration is the Weyl group $\mathbf{W(E_6) \cong O^-(6,2).2 \cong GO^-(6,2)}$, of order **51,840**. (Tue) Its simple subgroup of index 2 is $PSU(4,2) \cong PS\Omega^-(6,2) \cong PSp(4,3)$, of order **25,920**.

The dictionary between $F_2$ geometry and the 27-line configuration is remarkably complete:

| $F_2$ structure on $Q^-(5,2)$ | Classical object | Count | Lie-theoretic meaning |
|---|---|---|---|
| Isotropic points | Lines on cubic surface | **27** | Weights of the 27-dim rep of $E_6$ |
| Non-isotropic points | Double-sixes | **36** | Positive roots / reflections of $E_6$ |
| Totally isotropic lines | Tritangent planes | **45** | — |
| 6-cliques in the Schläfli graph | Maximal tori structures | **72** | Roots of the $E_6$ root system |

The **Schläfli graph** srg(27,16,10,8) (Tue) has the 27 isotropic points as vertices, with adjacency defined by the bilinear form evaluating to 1 (corresponding to skew lines). Its complement (Wikipedia) srg(27,10,1,5) is the

collinearity graph of the **generalized quadrangle GQ(2,4)** (TUE) (Wikipedia) — the polar space of $Q^-(5,2)$ with 27 points and 45 lines, each line containing 3 points and each point on 5 lines. The Gosset polytope $\mathbf{2_{21}}$ realizes these 27 vertices in 6-dimensional Euclidean space with full $E_6$ symmetry, (Wikipedia) (arxiv) and its dual $\mathbf{1_{22}}$ has **72 vertices** matching the $E_6$ roots. (Grokipedia)

## The Fano plane generates octonions and $E_8$

The Fano plane PG(2,2) — **7 points, 7 lines, 3 points per line** (Wikipedia) (HandWiki) — is the multiplication table of the octonions. Each line $\{e_i, e_j, e_k\}$ with a chosen cyclic orientation gives $e_i \cdot e_j = e_k$ and $e_j \cdot e_i = -e_k$. (Wikipedia) (arxiv) There are exactly **480 valid octonion multiplication tables** (Theoryofeverything) (ways to orient the Fano plane to produce a normed division algebra), decomposing as 30 Fano structures × 16 sign choices. (Theoryofeverything) These 480 tables split into two mirror classes of **240 each** (Theoryofeverything) — and this 240 equals the number of roots in $E_8$.

The automorphism group of the Fano plane is $\mathbf{GL(3,F_2) \cong PSL(2,7)}$, of order **168**. (Wikipedia +3) This embeds as a finite subgroup of the compact Lie group $\mathbf{G_2 = Aut(\mathbb{O})}$, which has dimension **14**. (Wikipedia) The key relationship: GL(3,2) permutes the unoriented Fano incidence structure, while $G_2$ additionally preserves the oriented multiplication with signs. The fixed-point subalgebra of the order-3 triality automorphism of $\mathfrak{so}(8)$ is precisely $\mathfrak{g}_2$, (University of California, Rivers…) (Wikipedia) and under $G_2 \subset SO(7)$, the 7-dimensional representation restricts to the action on imaginary octonions $Im(\mathbb{O})$. (Wikipedia)

The $E_8$ root system's **240 roots in $\mathbb{R}^8$** (Wikipedia) can be constructed directly from octonions (Wikipedia) using the Fano plane. (University of California, Rivers…) In the integral octonion (Cayley integer) description, (Academia.edu) (Wikipedia) the 240 unit integral octonions are:

- $\mathbf{\pm e_a}$ for a = 0,1,…,7: giving **16 roots**
- $\mathbf{\frac{1}{2}(\pm 1 \pm e_i \pm e_j \pm e_k)}$ where {i,j,k} is a Fano line, with an even number of minus signs: giving **7 × 16 = 112 roots** (University of California, Rivers…)
- $\mathbf{\frac{1}{2}(\pm e_p \pm e_\gamma \pm e_r \pm e_s)}$ where {p,q,r,s} is a Fano line complement, with sign constraint: giving **112 roots**

The Fano plane directly controls which coordinates cluster together in the half-integer roots. When reduced modulo 2, the 240 roots (in ±pairs) map to **120 points** forming a quadric in PG(7,2), and sub-$E_8$ lattices correspond to maximal flats (copies of PG(3,2)) inside this quadric. (University of California, Rivers…) (ucr)

## From binary codes to exceptional lattices

The pathway from $F_2$ projective geometry to exceptional lattices passes through coding theory with crystalline precision.

The **Hamming code [7,4,3]** is defined by a 3×7 parity check matrix whose columns are *all 7 nonzero vectors* of $F_2^3$ (Micsymposium) — equivalently, all 7 points of PG(2,2). (Errorcorrectionzoo) Weight-3 codewords correspond to the 7 lines of the Fano plane; weight-4 codewords to their complements. (Ucdenver) Extending by an overall

parity check gives the **[8,4,4] extended Hamming code**, Wikipedia the unique doubly even self-dual binary code of length 8 Errorcorrectionzoo LiquiSearch with **16 codewords** (weight enumerator $x^8 + 14x^4y^4 + y^8$).

**Construction A** transforms this code into the $E_8$ lattice: Rwth-aachen +2 $\Lambda(C) = (1/\sqrt{2})\{x \in \mathbb{Z}^8 : x \bmod 2 \in C\}$. The resulting lattice is even, unimodular, LiquiSearch and has **240 minimal vectors** — the $E_8$ root system. Euromathsoc +2 Viazovska proved in 2016 that this lattice achieves optimal sphere packing in 8 dimensions. Wikipedia

The same construction scales up spectacularly. The **extended binary Golay code [24,12,8]** — self-dual, Wikipedia with **759 octads** forming the Steiner system S(5,8,24) and automorphism group the Mathieu group $\mathbf{M_{24}}$ Southernct of order **244,823,040** — feeds into Construction A to yield the **Leech lattice $\Lambda_{24}$**, Brown University the unique even unimodular lattice in 24 dimensions with no roots and kissing number **196,560**.

The connection between levels involves triality. The Leech lattice can be built from **three copies of the $E_8$ lattice** Brown University via the Turyn construction, paralleling how the Golay code can be built from three copies of the extended Hamming code. En Academic The three $E_8$ copies sit in the vector, left-spinor, and right-spinor representations of Spin(8) — precisely the three representations permuted by $D_4$ triality. The n-Category Café arxiv The chain is:

**Fano plane → [7,4,3] Hamming → [8,4,4] ext. Hamming → $E_8$ lattice →×3 (via triality)→ Leech lattice → Conway groups → Monster**

The Golay code's 24 coordinate positions embed as **24 special points** (Todd's configuration) in PG(11,2), where $M_{24}$ acts as a collineation group. Springer Any 5 of these 24 points span a PG(4,2), but specific octads of 8 points lie in common 7-flats — this projective-geometric characterization is exactly the Steiner system S(5,8,24).

## Triality builds the $D_4 \to E_6 \to E_7 \to E_8$ tower

The Lie algebra $\mathbf{D_4 = \mathfrak{so}(8)}$ of dimension **28** has Dynkin diagram symmetry group $S_3$ — the unique diagram with triality. Wikipedia Its three outer nodes correspond to the three inequivalent 8-dimensional representations: $\mathbf{8_v}$ (vector), $\mathbf{8_s}$ (positive spinor), **8_c** (negative spinor). Triality cyclically permutes these, Wikipedia and its connection to octonions is that the isotopy group of $\mathbb{O}$ is Spin(8), acting via these three representations on domain, codomain, and range of multiplication. Texas A&M University Wikipedia

The exceptional groups arise by adding triality-related representations to $D_4$, parametrized by the four normed division algebras: arxiv

| Division algebra A | dim(A) | tri(A) | Exceptional algebra | dim |
|---|---|---|---|---|
| $\mathbb{R}$ | 1 | 0 | $\mathbf{F_4}$ | **52** = 28 + 8 + 8 + 8 |
| $\mathbb{C}$ | 2 | $\mathfrak{u}(1)^2$ (dim 2) | $\mathbf{E_6}$ | **78** |
| $\mathbb{H}$ | 4 | $\mathfrak{su}(2)^3$ (dim 9) | $\mathbf{E_7}$ | **133** |
| $\mathbb{O}$ | 8 | $\mathfrak{so}(8)$ (dim 28) | $\mathbf{E_8}$ | **248** = 28+28+64+64+64 |

The Barton-Sudbery triality formula gives dim L(A,B) = dim(tri(A)) + dim(tri(B)) + 3·dim(A)·dim(B). For $E_8$ = L($\mathbb{O}$,$\mathbb{O}$): 28 + 28 + 3×64 = **248**. Wikipedia Infogalactic The decomposition under $D_4 \times D_4$ is explicit:

$$\mathfrak{e}_8 = \mathfrak{so}(8) \oplus \hat{\mathfrak{so}}(8) \oplus (8_v \otimes \hat{8}_v) \oplus (8_s \otimes \hat{8}_s) \oplus (8_c \otimes \hat{8}_c)$$ Wikipedia +2

In the $E_8$ Dynkin diagram (Bourbaki numbering: nodes 1-3-4-5-6-7-8 in a line with node 2 branching off node 4), nodes **{2,3,4,5}** form the $D_4$ subdiagram with node 4 as the trivalent branch point. The key decompositions of the 248-dimensional adjoint under maximal subgroups are:

- **Spin(16)**: 248 = **120** ⊕ **128** (adjoint + half-spinor) Project Euclid
- $\mathbf{E_7} \times$ **SU(2)**: 248 = (133,1) ⊕ (1,3) ⊕ (56,2) Wikipedia
- $\mathbf{E_6} \times$ **SU(3)**: 248 = (78,1) ⊕ (1,8) ⊕ (27,3) ⊕ ($\overline{27}$,$\overline{3}$) Wikipedia
- $\mathbf{F_4} \times \mathbf{G_2}$: 248 = (52,1) ⊕ (1,14) ⊕ (26,7)

The $F_4 \times G_2$ decomposition directly shows $G_2$ (the octonion automorphism group, dimension 14) sitting inside $E_8$, with the 7-dimensional $G_2$ representation tensored with the 26-dimensional $F_4$ representation accounting for 182 of the 248 dimensions.

## The magic square works uniformly over $F_2$

The Freudenthal-Tits magic square produces the same Dynkin types over *any* field, including $F_2$. Infogalactic +2 The critical fact: **over finite fields, all composition algebras of dimension ≥ 4 are split** Wikipedia (by Wedderburn's theorem). Over $F_2$, the ingredients are:

| Dimension | $F_2$ algebra | Elements | Classical analog |
|---|---|---|---|
| 1 | $F_2$ | 2 | $\mathbb{R}$ |
| 2 | $F_4$ = GF(4) (non-split) or $F_2 \times F_2$ (split) | 4 | $\mathbb{C}$ |
| 4 | $M_2(F_2)$ (split quaternions, only option) | 16 | $\mathbb{H}$ |
| 8 | $\mathbb{O}_s(F_2)$ (split octonions, only option) | 256 | $\mathbb{O}$ |

The Albert algebra over $F_2$ is $H_3(\mathbb{O}_s(F_2))$ — 3×3 Hermitian matrices over split octonions with dimension **27 over $F_2$** (3 diagonal $F_2$-entries + 3 octonionic off-diagonal entries = 3 + 3×8 = 27), identical to the dimension over any field. In characteristic 2, the Jordan product $A \circ B = (AB+BA)/2$ requires McCrimmon's quadratic Jordan algebra formulation (using $U_x(y) = xyx$ instead). The automorphism group is **$F_4(2)$** and the structure group preserving the cubic determinant form is related to **$E_6(2)$**. (ScienceDirect) (Wikipedia)

The hierarchy above PG(5,2) follows the Severi varieties, which correspond to the second row of the magic square:

| **Severi variety** | **Ambient space** | **$F_2$-points of ambient** | **Lie symmetry** |
|---|---|---|---|
| Veronese $v_2(P^2)$ | $P^5$ | 63 | $A_2$ |
| Segre $P^2 \times P^2$ | $P^8$ | 511 | $A_2 \times A_2$ |
| Grassmannian Gr(2,6) | $P^{14}$ | 32,767 | $A_5$ |
| **Cayley plane $\mathbb{O}P^2$** | **$P^{26}$** | **134,217,727** | **$E_6$** |

The next level above PG(5,2) in the exceptional hierarchy is the **Cayley plane variety** ($E_{6,1}$ flag variety) embedded in PG(26,2), with $F_4(2)$ acting as automorphisms and $E_6(2)$ as the structure group. (Wikipedia) Beyond that, the Freudenthal variety lives in PG(55,2) with $E_7$ symmetry, and the $E_8$ geometry occupies PG(247,2).

## The 28 = 27 + 1 relationship reflects $E_7 \to E_6$ branching

The tantalizing near-coincidence **28 = 27 + 1** between off-Klein-quadric points and lines on a cubic surface is not accidental, but its explanation involves *two different quadrics* rather than a simple subset relation.

The 28 off-quadric points of $Q^+(5,2)$ and the 27 isotropic points of $Q^-(5,2)$ both live in PG(5,2) but correspond to different quadratic forms. The precise link comes from the **$E_7 \to E_6$ branching**: $E_7$ has a minimal 56-dimensional representation whose 56 weights pair into **28 pairs of opposite weights** (the 28 bitangents). Under the restriction $E_6 \subset E_7$, this representation decomposes as $56 = 1 \oplus 27 \oplus 27 \oplus 1^*$. The Weyl group $W(E_6)$ has **index 28** in $W(E_7)$ (arXiv) — choosing one of the 28 bitangent pairs as "special" breaks $E_7$ symmetry to $E_6$ and reveals the 27 lines. The **Gosset graph** (56 vertices, $E_7$ symmetry) has the Schläfli graph (27 vertices, $E_6$ symmetry) as its local graph at any vertex, (TUE) (Tue) geometrically expressing how each bitangent "sees" the 27 lines.

## Finite exceptional groups over $F_2$ and their representations

The Chevalley groups over $F_2$ are among the largest known finite simple groups, (Wikipedia) acting on $F_2$-vector spaces whose dimensions match the fundamental representations: (Wikipedia)

| Group | Order (factored) | Minimal rep. dim | Acts on |
|---|---|---|---|
| $G_2(2)$ (not simple) | $2^6 \cdot 3^3 \cdot 7 =$ **12,096** Wikipedia | 6 | Split octonions/$F_2$ |
| $F_4(2)$ | $2^{24} \cdot 3^6 \cdot 5^2 \cdot 7^2 \cdot 13 \cdot 17 \approx$ **$3.3 \times 10^{15}$** clemson | 26 | Traceless Albert algebra/$F_2$ |
| $E_6(2)$ | $2^{36} \cdot 3^6 \cdot 5^2 \cdot 7^3 \cdot 13 \cdot 17 \cdot 31 \cdot 73 \approx$ **$2.1 \times 10^{23}$** clemson | 27 | Albert algebra/$F_2$ with cubic form |
| $E_7(2)$ | $2^{63} \cdot 3^{11} \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 31 \cdot 43 \cdot 73 \cdot 127 \approx$ **$8.0 \times 10^{36}$** | 56 | Freudenthal triple system/$F_2$ |
| $E_8(2)$ | $2^{120} \cdot 3^{13} \cdot 5^5 \cdot 7^4 \cdot 11^2 \cdot 13^2 \cdot 17^2 \cdot 19 \cdot 31^2 \cdot 41 \cdot 43 \cdot 73 \cdot 127 \cdot 151 \cdot 241 \cdot 331 \approx$ **$3.4 \times 10^{74}$** | 248 | Adjoint (Lie algebra $e_8/F_2$) |

Each group acts on the $F_2$-points of its Tits building — a simplicial complex encoding the incidence geometry of the flag variety. For **$E_6(2)$**, the building has 6 types of vertices (one per Dynkin node), Wikipedia and its chambers (maximal simplices) number $|E_6(2)|/|B(F_2)|$ where B is a Borel subgroup. The "points" of the $E_6$ building are rank-1 elements of the Albert algebra over $F_2$, forming the Cayley plane variety. Wikipedia The number of chambers of the $E_6(2)$ building is $(2^{12}-1)(2^9-1)(2^8-1)(2^6-1)(2^5-1)(2^2-1) =$ **3,126,110,146,560**.

$G_2(2)$ is exceptional among these: it is **not simple**, HandWiki +2 with derived subgroup $G_2(2)' \cong PSU(3,3)$ of order 6,048. Wikipedia Over $F_2$, the Tits group ${}^2F_4(2)'$ of order **17,971,200** also appears — sometimes counted as a 27th sporadic simple group. Wikipedia +2

## Conclusion

The pathway from $F_2$ geometry to exceptional Lie groups follows a single architectural principle: **the combinatorics of subspaces over $F_2$ encode the weight structures and Weyl group actions of exceptional algebras**. The Fano plane PG(2,2) generates octonion multiplication Wikipedia and thereby $G_2$; Wikipedia +2 the elliptic quadric $Q^-(5,2)$ in PG(5,2) encodes the 27 lines and $W(E_6)$ as an orthogonal group $O^-(6,2)$; TUE Tue the Klein quadric $Q^+(5,2)$ encodes the 28 bitangents and $W(E_7)$ as a symplectic group Sp(6,2); arxiv and the full $E_8$ emerges from two copies of octonions through the magic square, Wikipedia with its 240 roots controlled by Fano plane combinatorics. arxiv The binary Hamming and Golay codes — both defined by $F_2$ projective geometry — build the $E_8$ and Leech lattices through Construction A, HandWiki Wikipedia linking coding theory to the deepest structures in algebra. EMS Press Euromathsoc The fact that $D_4$ triality underpins both the octonion-to-$E_8$ construction and the three-$E_8$-to-Leech construction reveals triality as the engine driving the entire exceptional hierarchy. Wikipedia Most remarkably, all of this works uniformly over $F_2$ itself: the magic square produces split Chevalley groups, Infogalactic the Albert algebra retains dimension 27, Ugent and the Tits

buildings provide finite geometric incarnations of the same incidence structures that govern continuous exceptional symmetry.